GLOBAL SCHOOL OF SCIENCE

First Terminal Examination - 2079

Subject: Mathematics (0071)

Class : XI | **Full Marks: 75**

SET 'B'

Time: 3 hrs. | **Pass Marks: 30**

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative. **[1 × 11 = 11]**

1. If $A \cap B = \phi$ and $x \in A$ implies that

   a) $x \in B$ b) $x \notin B$

   c) $x \notin \overline{B}$ d) $x \notin A \cup B$

2. The total number of subsets of the set B = {1, 2, 3} is

   a) 4 b) 8 c) 16 d) 32

3. A compound statement which is always true whatever may be the truth value of it's component is called

   a) tautology b) conjunction

   c) logically equivalent d) contradiction

4. The contrapositive of the compound statement $p \Rightarrow q$ is

   a) $\sim p \Rightarrow \sim q$ b) $\sim q \Rightarrow \sim p$

   c) $p \wedge (\sim q)$ d) $q \Rightarrow p$

5. The symmetric difference $A \Delta B$ of two sets A and B is defined as

   a) $(A - B) \cup (B - A)$ b) $(A \cup B) \cap (A \cap B)$

   c) $(A - B) \cap (B - A)$ d) both (a) & (b)

6. If A is a square matrix, then which of the following is always a skew-symmetric matrix?

a) $A + A^T$ b) $A - A^T$

c) $A A^T$ d) A^2

7. For what value of x the matrix A= $\begin{pmatrix} 6 & x-2 \\ 3 & x \end{pmatrix}$ is singular?

a) 3 b) 2 c) 0 d) – 2

8. If $\begin{pmatrix} x-y & x+y \\ y & x \end{pmatrix}$ is a upper triangular matrix with determinant 9, then the value of x is

a) 0 b) ± 1 c) ± 2 d) ± 3

9. The value of $\lim\limits_{x \to 0} \frac{\sin x^g}{x}$ is

a) 0 b) 1 c) $\frac{\pi}{200}$ d) $\frac{200}{\pi}$

10. $\lim\limits_{x \to -1} (x + x^2 + x^3 + \cdots\cdots\cdots + x^{10})$ is

a) 0 b) 1 c) –1 d) 2

11. The limit of the sequence $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}$ ……. is

a) 0 b) $\frac{3}{2}$ c) 2 d) does not exist

Group 'B'

Short Answer Questions. **[5 × 8 = 40]**

12. Define conditional statement with one example. Construct the truth table for the compound statement $(p \Rightarrow q) \Leftrightarrow (\sim q \Rightarrow \sim p)$. What conclusion can be drawn about the statement from truth table? [1+3+1]

13. a) If A and B are subsets of U, prove that $A - B = \overline{B} - \overline{A}$.

b) For any non-empty subsets A, B, C of universal set U, prove that:

$A - (B \cap C) = (A - B) \cup (A - C)$ [2+3]

14. a) If $A - 2B = \begin{pmatrix} 2 & 4 \\ -6 & 1 \end{pmatrix}$ and $A + B = \begin{pmatrix} 3 & 1 \\ -5 & 2 \end{pmatrix}$, find the matrices A and B.

b) If $P = \begin{pmatrix} 2 & 4 \\ -6 & 1 \end{pmatrix}$ and $Q = \begin{pmatrix} 3 & 1 \\ -5 & 2 \end{pmatrix}$ then verify that

$(P + Q)^T = P^T + Q^T$ and $(PQ)^T = Q^T P^T$. [2+3]

15. Without expanding the determinant prove that

a) $\begin{vmatrix} 3 & -15 & 1 \\ 2 & -10 & 5 \\ -5 & 25 & 1 \end{vmatrix} = 0$

b) $\begin{vmatrix} yz & x & x^2 \\ zx & y & y^2 \\ xy & z & z^2 \end{vmatrix} = \begin{vmatrix} 1 & x^2 & x^3 \\ 1 & y^2 & y^3 \\ 1 & z^2 & z^3 \end{vmatrix}$ [2+3]

16. If $A = \begin{pmatrix} 3 & -2 & 1 \\ -1 & 1 & 0 \\ 2 & -2 & -1 \end{pmatrix}$

i. Find the cofactors of each element of A. [2]

ii. Find adjoint of A. [1]

iii. Does the inverse of matrix A exist? If exists, find it. [1+1]

17. a) Define indeterminate forms. Evaluate $\lim_{x \to 3} \frac{x^2-7x+12}{x^2+2x-15}$

b) Evaluate: $\lim_{x \to \infty} (\sqrt{x+a} - \sqrt{x})$ [3+2]

18. a) Evaluate: $\lim_{x \to 1} \frac{x - \sqrt{2-x^2}}{2x-\sqrt{2-2x^2}}$

b) Evaluate: $\lim_{x \to a} \frac{x^2 - a^2}{\tan (x - a)}$ [3+2]

19. a) Evaluate: $\lim\limits_{x \to 0} \frac{\tan 2x - \sin 2x}{x^3}$

b) Evaluate: $\lim\limits_{x \to 0} \frac{1-\cos px}{1-\cos qx}$ [3+2]

Group 'C'

Long Answer Questions. **[8 × 3 = 24]**

20. a) Write the converse, inverse, negation and contrapositive along with their truth values of the statement, "If 6 is not an odd number, then 9 is an odd number." [4]

b) Define compliment of a set. If A and B be two subsets of a universal set U then prove that: $\overline{A \cap B} = \bar{A} \cup \bar{B}$. Also, if $A \cap B = \phi$ then prove $A \subseteq \bar{B}$. [1+1.5+1.5]

21. Prove that:

a) $\begin{vmatrix} p+x & p & p \\ q & q+y & q \\ r & r & r+z \end{vmatrix} = xyz\left(1 + \frac{p}{x} + \frac{q}{y} + \frac{r}{z}\right)$

b) $\begin{vmatrix} b+c & a & b \\ c+a & c & a \\ a+b & b & c \end{vmatrix} = (a+b+c)(c-a)^2$ [4+4]

22. a) Evaluate: $\lim\limits_{y \to x} \frac{y \sin x - x \sin y}{y - x}$ [4]

b) If $f(x) = \frac{ax+b}{x-5}$, $\lim\limits_{x \to 0} f(x) = -1$ and $\lim\limits_{x \to \infty} f(x) = 3$, find the value of f(6). [4]

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